(D) Generally Speaking (1/4) [20 Points]

Think about these two examples:

A. The hummingbird interviewed the penguin.

B. The the interviewed penguin hummingbird.

It's unlikely that you have seen either of these examples before. Nonetheless, you can probably tell that example A is a possible English sentence (it's certainly strange, but it is **grammatical**—that is, it obeys all the rules of English sentence structure), while example B could not be an English sentence.

How are we able to make judgments about sentences we have never seen before? This question relates to the broader topic of **generalization**—applying what you have learned to new situations or examples. Generalization is an important concept in computational linguistics because we want to build computers that can effectively generalize to sentences they have never seen before.

At NACLO Labs, to study generalization, we have created four different computers that are able to learn from data. The computers are named C1, C2, C3, and C4, and they have never encountered any English sentences before our experiment. We first show the computers the 17 sentences below, which we will refer to as the **training set**—the set of examples that the computers learn English from. We tell the computers that all 17 of these sentences are grammatical.

Training set:

1. The linguist met the programmer.
2. The cartoonist saw the spy.
3. The watchmaker is asleep.
4. The happy concierge met the ballerina.
5. The spy saw the woodcarver.
6. The spy is knowledgeable.
7. The main detective visited the ballerina.
8. The main calligrapher saw the astronaut.
9. The programmer visited the blacksmith.
10. The detective met the yodeler.
11. The spy is happy.
12. The ballerina is famous.
13. The linguist is tall.
14. The talented watchmaker visited the astronaut.
15. The famous programmer visited the blacksmith.
16. The cheerful concierge met the ballerina.
17. The haberdasher saw the linguist.

Note that all of the nouns in these sentences are types of occupations (for example, a haberdasher is someone who sells men's clothing), but for this problem you do not need to know the meanings of these words.

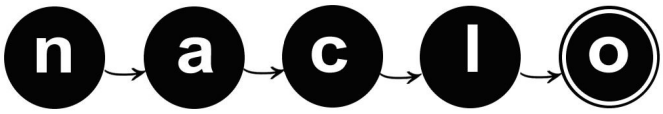
We now show the computers 21 more sentences, which are listed in the table on the next page. For each sentence, we ask each computer whether it thinks the sentence is **grammatical (G)** or **ungrammatical (U)**. Their answers are recorded in the table. We have also included the answers that a human provided when shown the same sentences.

(D) Generally Speaking (2/4)

D1. Some of the cells in the table are empty. On your Answer Sheets, provide the missing entries.

	Sentence	C1	C2	C3	C4	Human
18.	The main detective visited the ballerina.	G	G	G	G	G
19.	The linguist is tall.	G	G	G	G	G
20.	The cartoonist saw the spy.	G	G	G	G	G
21.	The linguist visited the spy.	U	G	G	G	G
22.	The main concierge saw the blacksmith.	U	G	G	G	G
23.	The ballerina is asleep.	U	G	G	G	G
24.	The cartoonist met the programmer.	U	G	G	G	G
25.	The woodcarver visited the programmer.	U	U	G	G	G
26.	The linguist saw the haberdasher.	U	U	G	G	G
27.	The famous concierge visited the watchmaker.	U	U	G	G	G
28.	The linguist is cheerful.	U	U	G	G	G
29.	The calligrapher watchmaker visited met.	U	U	U	G	U
30.	The knowledgeable cheerful talented.	U	U	U	G	U
31.	The tall detective saw the ballerina.	U	(a)	G	G	G
32.	The linguist saw the yodeler.	U	(b)	G	G	G
33.	The visited met linguist programmer.	U	(c)	U	G	U
34.	The yodeler met the woodcarver.	U	(d)	G	G	G
35.	The watchmaker is famous.	U	(e)	G	G	G
36.	The detective met the yodeler.	(f)	(g)	G	(h)	G
37.	The spy is talented.	(i)	(j)	G	(k)	G
38.	The programmer met the cartoonist.	(l)	(m)	G	(n)	G

<u>Make sure you record your answers in your Answer Sheets!</u>



(D) Generally Speaking (3/4)

D2. The table below contains some more sentences, along with C2 and C3's judgments for those sentences. In each sentence, one word has been hidden from you by replacing it with the string HIDDEN_WORD_1, HIDDEN_WORD_2, or HIDDEN_WORD_3. On your Answer Sheeet, write down what each HIDDEN_WORD stands for, choosing from the following options: ***asleep***, ***happy***, or ***main***. Additional notes:

- You should use each option (***asleep***, ***happy***, or ***main***) exactly once.
- Each HIDDEN_WORD stands for the same word in both of the sentences where it appears.

	Sentence	C2	C3
39.	The watchmaker is **HIDDEN_WORD_1**.	G	G
40.	The **HIDDEN_WORD_1** concierge visited the astronaut.	G	G
41.	The watchmaker is **HIDDEN_WORD_2**.	U	G
42.	The **HIDDEN_WORD_2** concierge visited the astronaut.	G	G
43.	The watchmaker is **HIDDEN_WORD_3**.	G	G
44.	The **HIDDEN_WORD_3** concierge visited the astronaut.	U	G

D3. For each sentence in the table above, think about what your own labels would be. Out of these 6 sentences, how many are there where C2 produced the same label as you? How many are there where C3 produced the same label as you? Write your answers on your Answer Sheet.

⇒ ***NOTE:*** Different people may have different judgments about these sentences. Therefore, as long as you write *something* in your Answer Sheet for **D3**, we will give you credit—it doesn't matter what you write! We just want you to think a bit about which computer matches your own judgments the best.

Ideally, we would want our computers to make exactly the same predictions as a human would. If you look back at the previous page (sentences 18 to 38), you will see that C1 and C2 often **undergeneralize**—there are sentences, such as sentence 25, that these computers label as ungrammatical (U) even though the human called them grammatical (G).

In the other direction, C4 sometimes **overgeneralizes**—it labels some sentences as grammatical (G) even though the human labeled them as ungrammatical (U). Depending on how you answered question **D3**, there may also be some examples of undergeneralization or overgeneralization in sentences 39 to 44 as well.

The examples of overgeneralization that we've seen so far were about the domain of sentence structure. Overgeneralization can also happen on the level of individual words. For example, based on pairs of words like *walk* and *walked*, or *laugh* and *laughed*, a computer might overgeneralize and produce the word *thinked* (based on *think*) or *bringed* (based on *bring*).

How might word-level overgeneralization affect our interactions with a computer? As an illustration, the next page provides some interactions between a human and a hypothetical robot. The robot has been trained on a very large amount of English text, and it often overgeneralizes when forming words. Each word that resulted from overgeneralization has been hidden from you by being encoded in a cipher, where each letter has been replaced with a different letter. The same code was used for all examples—for example, if **X** stands for w in the first encoded word, then it will also stand for w everywhere else.

(D) Generally Speaking (4/4)

D4. On your Answer Sheets, decode each encoded word from the conversations below. We are not providing the robot's training set, so it will not be possible to understand the robot as precisely as C1, C2, C3, and C4 from the previous part of the problem. However, if you carefully analyze what the robot says and also figure out the code, it should be possible to determine what each encoded word stands for.

Human: How severe will the storm be? Do you think it will interrupt my WiFi?

Robot: The storm will have a high level of severity. An interruption to your WiFi is likely, especially if lightning strikes somewhere nearby.

Human: Are you able to make this picture smaller for me?

Robot: I would be happy to **QGTRHUU** it for you! The **RKGKHENDKEW** of the smaller version should make it easier to move around. And I would be happy to enlarge it again later.

Human: That's a strange way to phrase it…but thank you! Also, something looks a little odd about the picture—can you tell what it is?

Robot: The picture is currently **UQYETKFQ-DKAPE**. I would be happy to apply a mirror filter to fix that. The website where you got the picture has many shortcomings—one of them is that it often distorts images.

Human: *[showing a picture]* What is the largest object in this picture?

Robot: *[pointing at object in picture]* That!

Human: Where is the water bottle in the picture?

Robot: *[pointing at corner of picture]* There!

Human: Who is the tallest person in the picture?

Robot: *[pointing at person in picture]* **EPX**!

Human: Can you attempt to schedule an appointment with my hairdresser?

Robot: Yes, I will make this **HEEQRMEKXG** soon. I will request an appointment **TXRQCPQG** next week.

Human: *[returning after a two-week vacation]* It's great to be back in the office! Did anything happen while I was away?

Robot: Not much happened for the first week. But then, **WQTEQDCQQB**, we got two new clients! They thought our company had many **EHUUJXRKGAT**, so they were happy to hire us.

An important challenge in computational linguistics is figuring out how to get computers to generalize in the right way. Generalizing can also be a challenge for humans acquiring languages. For example, when acquiring English, children often overgeneralize by producing words such as *thinked* or *bringed.*

Make sure you record your answers in your Answer Sheets!